Recommendations for estimating costs of sustaining maternal and neonatal tetanus elimination

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Background

Achieving Maternal and Neonatal Tetanus Elimination

Maternal and neonatal tetanus (MNT) elimination is defined as less than one case of neonatal tetanus per 1000 live births in every district of a country. In 1989 the 42nd World Health Assembly called for the elimination of neonatal tetanus by 1995. The 1990 World Summit for Children stated elimination of neonatal tetanus as one of its goals; this was endorsed by the 44th World Health Assembly in 1991. However, progress to attain elimination was slow and in 1999, the United Nations Children's Fund (UNICEF), World Health Organization (WHO) and United Nations Population Fund (UNFPA) re-launched the Maternal and Neonatal Tetanus Elimination (MNTE) Initiative to renew the goal of eliminating both neonatal tetanus and maternal tetanus. The main strategies involved in MNTE are the provision of tetanus toxoid (TT) vaccination through routine and supplementary immunization activities (SIAs) to vaccinate women of reproductive age (WRA) in areas with limited access to health services, strengthening clean birth services, and effective surveillance to detect areas and populations with high risk of neonatal tetanus (1).

Sustaining MNTE

Once MNTE has been achieved, sustaining elimination requires a comprehensive multi-pronged approach includes a complementary set of strategies focusing on the long-term protection against tetanus[1]. While these activities still focus on women, they expand and shift towards adjusting the national immunization schedule to provide long-term protection against tetanus for all people[1]

Table 1. Activities to achieve and sustain maternal and neonatal tetanus elimination (MNTE)

Achieving MNTE	Sustaining MNTE
Strengthening antenatal care (ANC) immunization of pregnant women with tetanus toxoid-containing vaccine (TTCV)	Strengthen immunization of pregnant women and routine vaccination of all children/ adolescents (both sexes) to receive 3 primary infant doses and 3 booster doses of TTCV before adolescence
TTCV Supplementary Immunization Activities (SIAs) in selected high-risk areas, targeting women of reproductive age (15-49 years) with 3 properly-spaced doses of the vaccine	Antenatal screening of pregnant women to verify tetanus vaccination status (to ensure tetanus protection at birth – PAB) and vaccinate if required
Promotion of clean birth and clean cord care practices and health education	Increased access to skilled health personnel at birth and clean birth/cord care practices
Reliable NT surveillance including case investigation and response	Strong tetanus/neonatal tetanus surveillance and regular review of data to identify districts at risk of re-emergence of MNT and needing corrective action

Approach

UNICEF established an ad hoc expert consultative group to provide advice on the assumptions and inputs used to estimate the cost of four interventions required to sustain MNTE: tetanus-diphtheria (Td) vaccination of pregnant women during antenatal care (ANC); tetanus vaccine booster at second year of life and two school-age booster doses; skilled delivery including clean delivery and cord care; and reinforcement of MNT surveillance and monitoring sustainability of MNTE. An ingredients-based costing approach was used to estimate the financial cost of providing these intervention; the approach followed the methodology used in the Lives Saved Tool Costing module [2]. This approach estimates the target population on which the health intervention is focused, the population in need of the intervention, the intervention coverage, the treatment inputs (i.e., drugs and supplies, medical personnel time, and number of inpatient days and outpatient visits) per case, and the costs per service. Facility level costs and above-facility (or program) costs necessary to support the intervention were also included. The expert consultative group also provided guidance to estimate the cost of activities needed to reinforce MNTE surveillance and monitor MNTE sustainability in terms of their frequency and unit costs.

Original costs were captured in 2018 USD. They have been adjusted for inflation to 2026 using https://www.usinflationcalculator.com/ and are reported in both USD and CAD.

Methods to estimate the cost of Td vaccination, tetanus vaccine boosters, and skilled delivery including clean delivery and cord care

Estimating the cost of Td vaccination during ANC

Properly-spaced doses of a TTCV during the antenatal period can reduce the incidence of neonatal tetanus by 88% to 100%[3]. Achieving and sustaining MNTE requires that countries ensure that at least 80% of pregnant women in every district are fully vaccinated against tetanus[1]. Since it will take time to achieve this, the only way to protect all mothers and newborns from tetanus is to screen pregnant women during ANC to check their tetanus vaccination status, and to administer any TTCV doses needed for long-term protection. As of January 2020, countries have transitioned from the tetanus toxoid vaccine to the tetanus-diphtheria (Td) vaccine which ensures dual protection from tetanus and diphtheria[4]. Cost data inputs, assumptions and sources are described below and in Table 1.

Notes:

- Population in need of Td vaccination during ANC captures all women having a first and second pregnancy, the women who are not covered by Td during ANC (i.e., 100% coverage minus current coverage levels), as well as those women who had three more pregnancies who are already protected by having received two doses of Td during ANC. Calculation:
  (% women uncovered by ANC4 x % women having a 3rd or greater birth) + women having a 1st or 2nd birth
- Instead of using TT2+ coverage (i.e., the proportion of pregnant women who have received their second or superior TT dose in a given year), the expert consultative group suggested using ANC4 coverage (i.e., the percentage of pregnant women that received at least four ANC visits during a specified time period) since it is recognized that TT2+ coverage is likely an under-estimate in most

countries because most pregnant women do not have records of the TTCV doses they have received during previous pregnancies.

- Labor cost for a midwife to provide Td vaccination is included, although it is acknowledged that this might be a shared cost with other programs using the ANC platform to deliver interventions, and might be a recurrent expenditure as part of broader efforts to strengthen health systems.

Table 1. Td during ANC: baseline data Inputs, unit costs and sources

Input	Description
Target population	# of pregnant women
Population in need (%)	Proportion of women in need of two Td vaccinations during ANC (i.e., pregnant women not reached through ANC4 and those women experiencing a first or second pregnancy)
Baseline Coverage	ANC4 coverage
Td vaccine unit (dose) price	Td 10 dose vial + needle/syringe/swab[5] USD $0.51 (CAD $0.71)
Td vaccine wastage	25%[6]
Labor	5 minutes of time for a midwife[7] 2 outpatient visits[8]

Estimating the Cost of Clean and Skilled Delivery[1]

Cutting of the umbilical cord at birth can be a pathway for local and invasive infections, which have a high case-fatality rate if untreated. A clean birth, followed by appropriate hygienic practices post-childbirth, can reduce the risk of maternal and neonatal tetanus and sepsis. Having skilled health personnel is key to ensuring clean birth and cord care, and skilled birth assistance has proven to be an effective intervention to reduce neonatal tetanus[9–11]. Presence of a skilled birth attendant during delivery ensures that critical components of clean delivery practices, including clean surfaces, clean hands, and clean cord cutting instruments, are used during the delivery; it also presents an opportunity for health workers to communicate clean cord care practices to the mother, since unclean cord-care practice is a significant risk factor for neonatal tetanus. The WHO has endorsed the supply of clean delivery kits in resource-poor settings for many decades to increase uptake of clean birth practices; the contents of the clean delivery kit differ from country to country depending on local guidelines and regulations, as well as on where it is intended to be used[12].

The application of potentially harmful substances on the umbilical stump by many populations, often due to lack of knowledge about proven methods of sanitary cord care, or cultural beliefs on the need to apply something to the cord to facilitate healing of the stump, are a significant contributor of neonatal tetanus, particularly for home births[13]. Chlorhexidine (7.1% chlorhexidine digluconate aqueous solution or gel, delivering 4% chlorhexidine) is a broad-spectrum topical antiseptic with a residual effect for up to 72 hours. Evidence from community-based randomized controlled trials in Bangladesh, Nepal and Pakistan demonstrated the impact of chlorhexidine application to the umbilical cord stump in preventing NT infections and neonatal sepsis[14]. WHO has added chlorhexidine to its Model List of Essential Medicines for Children, and issued recommendations for daily chlorhexidine application to the

[1] Note: it is recognized that while clean and skilled delivery is a key strategy to sustain MNTE, it has and has broader implications beyond MNTE. Countries can use the inputs and assumptions here if they wish to estimate these costs.

umbilical cord stump during the first week of life for newborns who are born at home in settings with high neonatal mortality (i.e., 30 or more neonatal deaths per 1000 live births)[15].

Cost data inputs, assumptions and sources are described in Table 2 and Table 3.

Table 2. Chlorhexidine

Input	Description
Target population	# of births in 8 countries
Population in need (%)	% home births
Chlorhexidine	Chlorhexidine digluconate, 10ml bottle[16] USD $0.56 (CAD $0.78)
Transportation costs	Cost of transporting chlorhexidine bottle to home: USD $0.87 (CAD $1.22)[13]

Table 3. Clean and skilled delivery at home and facility births

Input	Home birth	Facility birth
Target population	# of births	# of births
Population in need (%)	% home births	% facility births[17]
Clean delivery kit	Clean delivery kit US$ 1.84 (CAD $2.58) (soap, plastic sheeting, razor blade, umbilical tape, cotton wrap for newborn)[18]	
Baseline coverage	% skilled birth attendance (same rate was used for home and facility births due to data limitations)[19]	
Labor costs	None	30 min of an Obstetrician/Gynecologist 360 minutes of a Midwife [7] 1 inpatient visit[8]

Estimating the Cost of Tetanus Boosters

The 2017 updated WHO guidance on tetanus vaccination recommends that national immunization programs should provide a total of six doses, consisting of three primary infant doses and three booster doses at the ages of 12-23 months, 4-7 years, and 9-15 years [4].

Costs of 2YL tetanus booster

The second year of life (2YL) platform is a critical step in extending immunization beyond infancy; it offers an opportunity to integrate vaccination with other critical health interventions[20]. Many countries already provide routine vaccine doses to children in the 2YL or later, and this platform for vaccination is expected to increase in the coming years. The 2YL platform provides an important opportunity to reach children with a tetanus booster which, in addition to the primary series of three TTCV doses in infancy, provides 3-5 years of protection against tetanus[1]. This 2YL tetanus booster provides an opportunity for catch-up of vaccine doses for children who might have missed earlier doses, and it also aligns with other WHO childhood vaccination recommendations, including the second dose of measles-containing vaccine (MCV2), meningitis A, and boosters of pertussis and diphtheria[1].

WHO and UNICEF have developed guidance material to assist countries establish and strengthen immunization in the 2YL and beyond[21]. The costs of a tetanus booster in the 2YL was estimated from this guidance as well as the UNICEF 2YL Costing Tool, and raw data from the analysis "Estimating the costs of strengthening a second year of life healthy child visit in Namibia and Cape Verde" that used the 2YL Costing Tool[20, 21].

Notes:

- To estimate the additional costs associated with provision of the 2YL tetanus booster, we referred to the UNICEF analysis in Namibia; the country already had existing 2YL activities but wanted to determine the cost of adding new interventions to existing ones to better understand all costs – including operational and training costs for 5 years[20]. The Namibia analysis used demographic data, vaccine and commodity parameters, including prices, health personnel salaries and training and communication costs. Using this raw cost data, we stripped the non-tetanus commodity and equipment costs, and assumed that communication, training and staff time costs are the same regardless of the health package being delivered. These costs for communication, training and staff time included start-up and recurring costs on an annual and every 3-year basis.

These assumptions and inputs are detailed in

Table *2*.

Table 2. Tetanus booster in 2YL

Input	Description
Target population	2-year old boys and girls
Population in need (%)	100%
Coverage	Measles-Containing Vaccine Second-Dose (MCV2)
Td vaccine price + Needle/syringe/swab	Td 10 dose vial + needle/syringe/swab[5] USD $0.51 (CAD $0.71)
Vaccine wastage	25%[6]
Staff time	USD $1.65 (CAD $2.31) per child reached, annually[20]
Training	USD $0.27 (CAD $0.38) per child reached, every 3 years[20]
Communication	USD $0.35 (CAD $0.49) per child reached, every 3 years[20]

Costs of school-age tetanus booster doses

An additional booster dose in early childhood or at school entry (i.e., at 4-7 years), in addition to the primary series of three TTCV doses in infancy, protects children into adolescence; and a third TTCV booster dose during adolescence (i.e., 9-15 years) will induce immunity that lasts through much of adulthood, which will protect women through their childbearing years[1]. The school platform takes advantage of high enrollment rates, and can be integrated into a comprehensive package of school health interventions to share costs and benefits[22].

Notes:

- To simplify estimation of the population of school-age children to receive the tetanus boosters, we assumed that in each country boys and girls receive the booster doses at age 7 and 11.

- Coverage was assumed to be 80% based on literature reviewed and the recommendation of the MNT Consultative Group[23].
- There is little literature documenting the unit cost of school-based tetanus booster, thus we relied on a consultant to determine the unit cost of the school-age tetanus booster in Indonesia – an immunization program that is well established with coverage between 81-99% depending on the province. For other countries, a ratio based on the GDP in the country being estimated compared to Indonesia can be used to convert the unit cost to appropriate estimates for another country.
- For countries with no school-based vaccination program, start-up costs were estimated using an average of the start-up costs from school-based HPV programs in Vietnam and Uganda[24].

The assumptions and inputs to estimate the cost of two tetanus booster doses for school-age children are documented below, in **Table *3***.

Table 3. School-age tetanus booster: baseline data inputs, unit costs and sources

Input	Description
Target population	7- and 11-year old boys and girls
Population in need (%)	Primary school attendance (% net) defined as the number of children attending primary school who are of primary school age, expressed as a percentage of the total number of children of primary school age
Coverage (among those countries with existing school tetanus vaccination programs)	80%[23]
Td booster	USD $1.20 (CAD $1.69)
Start-up costs	0.56 ratio applied to incremental service delivery cost for 2 years to reflect costs of starting up school-based vaccination programs

Cost of reinforcement of MNT surveillance and monitoring sustainability of MNTE

Background

Surveillance and monitoring of the vaccine-preventable disease (VPD) tetanus requires "*the systematic, timely, and continuous collection, analysis and interpretation of epidemiological data on tetanus cases, linked with timely dissemination of the results so that appropriate action can be taken to control or prevent further cases of tetanus*"[1]. The key objective of neonatal tetanus surveillance and monitoring is to detect cases of neonatal tetanus in order to monitor and achieve maintenance of MNTE; the objective of non-neonatal tetanus surveillance is to monitor disease burden and changing epidemiology over time, to assess the impact of vaccination and identify gaps[25]. Since tetanus surveillance is integrated within VPD surveillance, it was not possible to obtain the exact cost of MNT surveillance activities. The expert consultative group worked with a group of MNT Surveillance Experts to identify key surveillance activities for which to estimate costs

Notes

- The economic costs needed to sustain MNTE were estimated; definition of these costs were provided by MNT Surveillance Experts[10] and the MNT Consultative Group.
- For countries identified to have weak vaccine programs, several activities to reinforce MNTE surveillance and monitor MNTE sustainability are needed.

- No costs were estimated for integrated tetanus surveillance review as part of assessment of maintenance of MNTE, as the assumption is that this should be integrated into periodic Expanded Programme on Immunization (EPI) review and VPD surveillance reviews.

The assumptions and parameters are documented below and in Table 5.

Table 5. Reinforcing MNTE surveillance and monitoring MNTE sustainability

Sub-Activity	Description	Frequency	Unit cost (per country)
Refresher training for surveillance officer and surveillance focal point	Integrated into routine refresher training; costs include share of training costs (venue, materials, trainer fees or time, participant per diem/transport, participant time)	Every 3-5 years	USD $52,000-$87,000 (CAD $73K-122K) depending on country size[2]
Sensitization of clinical staff	Sensitization of surveillance officers and clinical staff during routine supervision visits and meetings	Every 3-5 years	USD $52,000-$87,000 (CAD $73K-122K) depending on country size[3]
Active & Passive Surveillance			
Surveillance officer visits major health facilities	Costs include time for surveillance officer to review reports (assume 1-2 hours); assume travel costs integrated with Acute Flaccid Paralysis (AFP) and Measles/Rubella (MR) active surveillance	Annual	USD $13,000 to $70,000 (CAD $19,000 to $97,000) depending on country size[3] (for countries with weak vaccine programs[4])
Community-based surveillance			

[2] Country size based on total population 2020-2030. Small (population <10 million USD $52,000 (CAD $73,000); Small-medium (population 10-50 million) USD$61,000 (CAD $85,000); Medium-large (population 5-100 million) USD $70,000 (CAD $97,000); Large (population 100-200 million) USD $87,000 (CAD $122,000)

[3] Country size based on total population 2020-2030. Small (population <10 million USD $52,000 (CAD $73,000); Small-medium (population 10-50 million) USD$61,000 (CAD $85,000); Medium-large (population 5-100 million) USD $70,000 (CAD $97,000); Large (population 100-200 million) USD $87,000 (CAD $122,000)

[4] Defined as countries that still need to achieve MNTE and countries where PAB < 85%

Community sensitization	Surveillance officer or clinical staff sensitize communities and birth attendants on NT; assume travel costs integrated with Acute Flaccid Paralysis AFP and MR active surveillance	Annual	USD $13,000 to $70,000 (CAD $19,000 to $97,000) depending on country size[3] (for countries with weak vaccine programs[5])
Assessment and maintenance of MNTE			
Integrated annual desk review	This cost is typically integrated into existing annual program reviews and surveillance reviews	Annual	USD $2,700 (CAD $3,800)/country (accounts for share for tetanus surveillance)
Interim post-validation assessment of MNTE	Involves EPI, MCH and surveillance program staff, travel and meeting costs)	Every 4 years	USD $87,000 (CAD $122,000) (for countries with weak vaccine programs[6])

Evaluation:

These inputs and assumptions were used to estimate the costs, impact and cost-effectiveness for sustaining MNTE in the 59 priority countries under four scenarios: a status quo scenario assuming intervention coverage will remain unchanged, and three scenarios assuming different intervention coverage scale-up.

Implications:

MNTE attainment and sustainability require interventions and strategies beyond the immunization program. In addition to supporting evidence-based planning, a standardized methodology can provide clarity on the costs of these important strategies, and recognition that countries will need to continue to invest in maternal and newborn health as part of their MNTE sustainability efforts.

[5] Defined as countries that still need to achieve MNTE and countries where PAB < 85%

[6] Defined as countries that still need to achieve MNTE and countries where PAB < 85%

References:

1. (2019) Protecting all against tetanus. Guide to sustaining maternal and neonatal tetanus elimination (MNTE) and broadening tetanus protection for all populations. World Health Organization, Geneva

2. Bollinger LA, Sanders R, Winfrey W, Adesina A (2017) Lives Saved Tool (LiST) costing: a module to examine costs and prioritize interventions. BMC Public Health 17:782

3. Roper MH, Vandelaer JH, Gasse FL (2007) Maternal and neonatal tetanus. The Lancet 370:1947–1959

4. (2018) Tetanus vaccines: WHO position paper, February 2017 – Recommendations. Vaccine 36:3573–3575

5. UNICEF Supply Hub. https://supplyhub.unicef.org/. Accessed 25 Oct 2026

6. World Health Organization (2020) Training for mid-level managers (MLM): Module 1 : cold chain, vaccines and safe-injection equipment management. World Health Organization

7. Evans DB, Edejer TT-T, Adam T, Lim SS (2005) Methods to assess the costs and health effects of interventions for improving health in developing countries. BMJ 331:1137–1140

8. World Health Organization (2021) WHO-CHOICE estimates of cost for inpatient and outpatient health service delivery.

9. Darmstadt GL, Bhutta ZA, Cousens S, Adam T, Walker N, De Bernis L (2005) Evidence-based, cost-effective interventions: how many newborn babies can we save? The Lancet 365:977–988

10. World Health Organization (2019) The global vaccine action plan 2011-2020: review and lessons learned: strategic advisory group of experts on immunization.

11. (2016) Global Routine Immunization Strategies and Practices (GRISP): a companion document to the Global Vaccine Action Plan (GVAP). World Health Organization, Geneva, Switzerland

12. (2015) WHO Safe Childbirth Checklist Implementation Guide: Improving the quality of facility-based delivery for mothers and newborns. World Health Organization, Geneva, Switzerland

13. Laing SK, Griffiths U, Raza AA, Zulu F, Yakubu A, Bessias S, Ozawa S (2020) An investment case for maternal and neonatal tetanus elimination. Vaccine 38:2241–2249

14. Mullany LC, Darmstadt GL, Khatry SK, Katz J, LeClerq SC, Shrestha S, Adhikari R, Tielsch JM (2006) Topical applications of chlorhexidine to the umbilical cord for prevention of omphalitis and neonatal mortality in southern Nepal: a community-based, cluster-randomised trial. The Lancet 367:910–918

15. (2013) WHO Recommendations on Postnatal Care of the Mother and Newborn. World Health Organization, Geneva

16. Management Sciences for Health, World Health Organization (2016) International Medical Products Price Guide. MSH, Medford, Mass

17. Indicator Metadata Registry Details. https://www.who.int/data/gho/indicator-metadata-registry/imr-details/institutional-birth. Accessed 26 Oct 2026

18. UNICEF Supply Hub. In: Clean delivery kit. https://supplyhub.unicef.org/product-detail?q=G5BzVTb3bxanJxUbuvLkRYJaaXkidUxuBu9DrXG3RVtJmnIOVq9jZ0oOtnKmVt4t. Accessed 25 Oct 2026

19. Births attended by skilled health personnel (%). https://www.who.int/data/gho/data/indicators/indicator-details/GHO/births-attended-by-skilled-health-personnel-(-). Accessed 26 Oct 2026

20. (2019) Estimating the costs of strengthening a Second Year of Life (2YL) healthy child visit in Namibia and Cape Verde. Immunization Economics

21. World Health Organization (2018) Establishing and strengthening immunization in the second year of life: practices for vaccination beyond infancy. World Health Organization, Geneva

22. Robert Steinglass (2016) Broader tetanus prevention: Impact of current tetanus immunization programmes and need for acquiring broader tetanus prevention.

23. LaMontagne DS, Cernuschi T, Yakubu A, Bloem P, Watson-Jones D, Kim JJ (2017) School-Based Delivery of Vaccines to 5- to 19-Year Olds. Child and Adolescent Health and Development

24. Levin CE, Van Minh H, Odaga J, Rout SS, Ngoc DNT, Menezes L, Araujo MAM, LaMontagne DS (2013) Delivery cost of human papillomavirus vaccination of young adolescent girls in Peru, Uganda and Viet Nam. Bull World Health Organ 91:585–592

25. World Health Organization (2018) Neonatal Tetanus. World Health Organization